$$J(\mathbf{w}, b) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)} \right)^2 + \frac{\lambda}{2m} \sum_{j=1}^{n} w_j^2$$